termine the relationships of *Masripithecus* with other Miocene apes. Traditional phylogenetic methods rely on the principle of maximum parsimony, which seeks the simplest explanation for determining relationships by minimizing the number of evolutionary changes. By contrast, a Bayesian approach uses a different, probabilistic method that may incorporate the age of fossils (tip dating) into the phylogenetic analysis to estimate divergence times (*12*). The authors' tip-dating analysis supported the stem hominoid status of presently recognized extinct ape lineages. However, it placed *Masripithecus* as the stem hominoid that is most closely related to crown hominoids.

Al-Ashqar *et al.* did not analyze the phylogenetic relationship between *Masripithecus* and *Nacholapithecus* from Kenya (15 Ma), which has been interpreted as either a stem hominoid (*10*) or a hominid (*13*). Moreover, the authors' maximum-parsimony and Bayesian analyses that did not incorporate the age of fossils resulted in different phylogenetic tree topologies than the tip-dating analysis, as they placed *Masripithecus* as a crown (instead of a stem) hominoid. Future phylogenetic analyses that include *Nacholapithecus* and use other datasets are required to conclusively discount a crown hominoid status for *Masripithecus*.

Because most known remains of stem great apes are restricted to East Africa (*1*, *3*), fossils from elsewhere in this continent are very valuable to reconstruct Miocene ape evolution. *Masripithecus* is particularly relevant because it indicates the presence of apes that seem closely related to crown hominoids in North Africa. This suggests that crown hominoids might have originated and diversified during the Early Miocene in northeastern Afro-Arabia, with various lineages dispersing into Eurasia after the establishment of land corridors ~20 to 19 Ma. The geographic bias of the fossil record might explain why the ancestral African stocks of hylobatids and other Eurasian primate groups, such as pliopithecoids (*14*), have yet to be identified. Three decades ago, the paleoanthropologist David Pilbeam referred to the "drunk's dilemma" regarding previous Miocene ape research, noting that paleoanthropologists "were like the drunk looking for his keys under the lamppost where it was light rather than where he had dropped them" (*15*). Al-Ashqar *et al.*'s findings on *Masripithecus* confirm that paleontologists might have been looking for crown-hominoid ancestors in the wrong place. This lesson should be taken with humility but also with renewed optimism, as the required fossils to complete the puzzle of Miocene ape evolution may be somewhere out there, waiting to be discovered. □

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ENVIRONMENTAL SCIENCE

Cascading impacts of natural disasters in a connected world

Multifaceted effects of extreme climate-related events are essential for risk models and management

Laurie S. Huning[1] and **Manuela I. Brunner**[2,3,4]

In 2023, extreme heat propelled the spread of severe wildfires across Canada, resulting in record burned area and smoke plumes that crossed international borders. Such massive fires can have broad consequences, including loss of human life, property, livelihood, flora, and fauna. The emitted aerosols worsen air quality, which can cause health problems (*1*). When deposited on snow and glaciers, the aerosols can intensify melt patterns and alter peak streamflow timing, causing downstream effects on water supply (*2*). Moreover, they affect the climate by absorbing and scattering solar radiation (*1*). Wildfires illustrate how extreme natural disasters can trigger cascades of events that have local and distant impacts across sectors. Although the interconnectivity among these occurrences is becoming more important with global warming, the relationship is often not well described, monitored, or represented in hazard models. This hinders accurate estimations of damages and effective preventative strategies.

Cascades of consequences from natural disasters can be seen in numerous examples around the world. In the case of the 2023 Canadian wildfires, smoke degraded air quality in the United States, including major metropolitan areas such as Chicago, Detroit, Minneapolis, Washington, DC, and New York City (see the figure). Consequently, these cities ranked among the 10 major cities with the worst air quality in the world in 2023 (*3*). The smoke was carried across the Atlantic Ocean as well and was even estimated to have contributed to more than 20,000 chronic deaths in Europe (*4*).

The effects of severe drought serve as another example and are readily observed in stressed vegetation, physical and biological changes to soil, and degradation of earthen infrastructure such as foundations, levees, and roads. Reduced soil moisture then leads to less evapotranspiration—the movement of water from Earth's surface to the atmosphere through evaporation from land and water, and transpiration from plants, which transport water into the atmosphere. This decreases atmospheric moisture that is available for precipitation and contributes to increased temperatures and the self-propagation of drought and subsequent heat waves (*5*).

The impact cascade of extreme drought and flooding in agricultural regions affects international trade and population-level dietary health. Countries that rely on imports for certain crops may suffer from price increases due to shortages. For example, the 2010 drought in Russia reduced its wheat yield, which led to a ban on wheat exports. Coupled with simultaneous wheat production deficits in other

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countries (for example, also caused by drought in China and western Australia, and cool and wet conditions in regions of Canada), this contributed to higher prices for wheat flour and bread. This event in Russia exacerbated social inequality and poverty in distant regions such as Egypt and Mozambique (*6*). In 2023, India, which is a major exporter of rice, imposed an export ban on specific varieties of rice after heavy precipitation and flooding devastated crops. The range of consequences disproportionately affected low-income consumers across the globe that relied heavily on rice imports as prices surged.

Although natural disasters may appear to only have local effects in nonagricultural regions given their visible footprint, global ramifications are possible if an event affects critical infrastructure such as semiconductor chip factories, oil fields, and refineries. If a local event such as an extreme storm and flooding devastated semiconductor facilities, for example, massive inundation could disrupt manufacturing and the global supply chain. Thus, supplies may be limited, which would trigger substantial price increases. The loss of chips also could stall the repairs, operations, and technological advancements for electronic devices in a myriad of areas, including health care, transportation, communications, and finances. The cascading impacts of chip shortages were already seen across many industries around the world during the COVID-19 pandemic.

Discussions of cascading impacts have largely been described as a chain of events analogous to toppling dominoes (*7*, *8*). However, a critical component of understanding these cascades lies in observing and modeling their role in feedback loops or wave-like actions with the enhancement of hazards or impacts. Wildfires, for example, increase the likelihood of heat waves through the loss of forest cover, corresponding increases in surface temperatures, and reductions in evapotranspirative cooling, which create conditions that favor future fires (*9*). During drought, soil desiccation cracking can expose carbon stored in soil to oxidation, which releases greenhouse gases from the soil into the atmosphere (*10*). Increased greenhouse gas concentrations in the atmosphere form a positive feedback loop that accelerates climate change (*10*). This could enhance drought conditions and drive widespread effects throughout the global system.

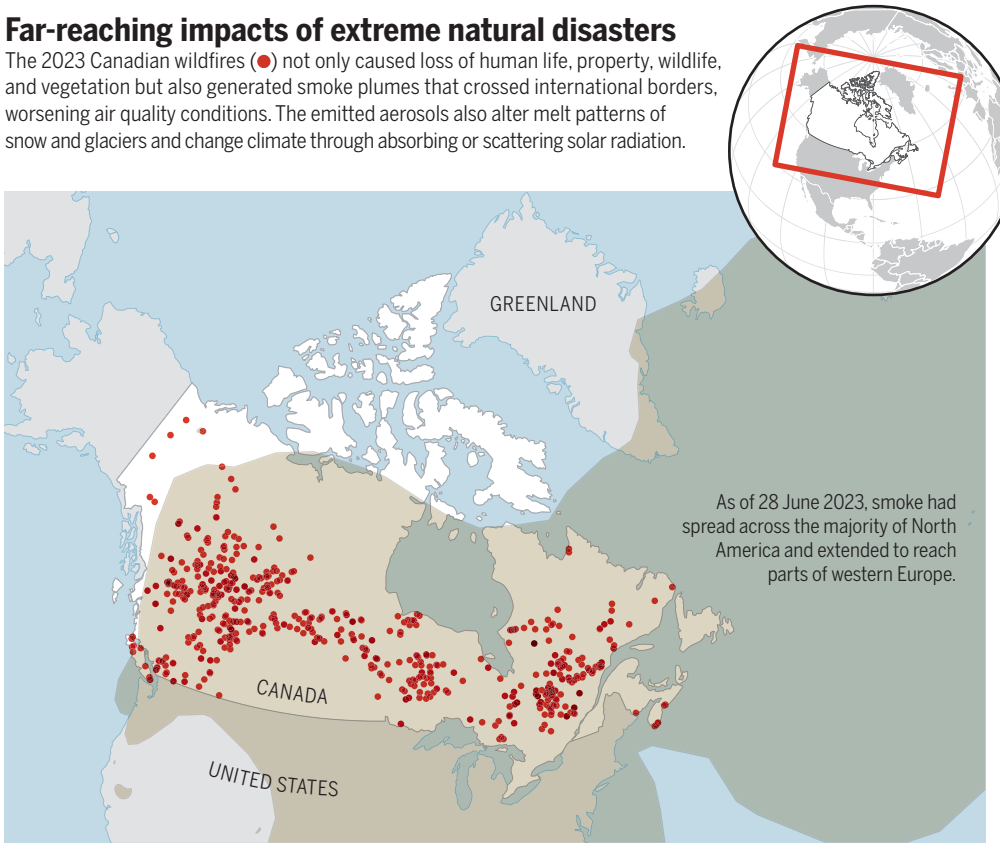
Extreme heat in France in summers 2024 and 2025 created another type of feedback loop. The rising air temperatures have caused substantial increases in river temperatures across the country. Water from rivers flowing through France, such as the Rhône and Garrone Rivers, is commonly used in France and neighboring countries to cool reactors at nuclear power plants. Regulations in the country on major power plants serve as safeguards to prevent the discharge of water that is too hot for downstream aquatic ecosystems to thrive. Hot river temperatures can cause insufficient cooling of nuclear power plants, reducing the amount of electricity generated. Such increases in river temperatures usually occur during heat waves, coinciding with additional strain on the power grid from cooling systems (such as air conditioning) and the limited efficiency of solar panels as an energy source. These issues are expected to be exacerbated because heat wave frequency, intensity, and duration are projected to increase across the country (*11*). Although France traditionally exports excess power to neighboring countries, a shift to importing power would reshape the European energy market. Increases in fossil fuel power would contribute to increased greenhouse gas emissions and air pollution, leading to higher temperatures, more energy consumption, and potentially additional fossil fuel emissions—a positive feedback. Thus, both local and remotely propagating implications of events must be considered in natural disaster impact models and assessments.

Human intervention such as forest or water management and infrastructure development can mitigate the effects of natural disasters. For example, removal of excess fuel for fire through proactive forest management reduces wildfire risk (*12*). Nevertheless, human response and adaptation strategies to one extreme such as drought could have unintended consequences affecting another hazard such as flooding (*13*). In addition, increased groundwater pumping during a drought often compacts aquifer sediments, causing land subsidence (*10*). Although subsidence can be recoverable or unrecoverable, overdraft of a groundwater aquifer could permanently reduce its potential water storage capacity and contribute to future drought conditions (*14*). This relative decline of land surface elevations makes the region more susceptible to flooding (*15*). Indeed, land subsidence exacerbated the flooding experienced in Houston, Texas, from Hurricane Harvey in 2017 and contributed to the levee failures during Hurricane Katrina in New Orleans, Louisiana, in 2005 (*10*, *13*, *15*).

Furthermore, monitoring and modeling feedbacks of impact cascades poses challenges because they commonly involve nonlinearities and multiple events at a given location, which are difficult to disentangle. Cascading effects also occur at large spatial and temporal scales and are not contained within a physical boundary. They take place across many timescales. For example, although the onset and spread of a wildfire may happen suddenly, its effects could take place at shorter timescales (such as air quality declines), longer timescales

Far-reaching impacts of extreme natural disasters

The 2023 Canadian wildfires (●) not only caused loss of human life, property, wildlife, and vegetation but also generated smoke plumes that crossed international borders, worsening air quality conditions. The emitted aerosols also alter melt patterns of snow and glaciers and change climate through absorbing or scattering solar radiation.



CREDITS: (GRAPHIC) V. PENNEY/*SCIENCE*; (DATA) CANADIAN INTERAGENCY FOREST FIRE CENTRE, INC.; NOAA HAZARD MAPPING SYSTEM FIRE AND SMOKE PRODUCT

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(such as forest regrowth and recovery), or a combination of both. Moreover, society usually focuses on quantifying direct economic damage from natural disasters, whereas other effects such as social inequalities, mental health issues, and human migration go unnoticed. As a result, a more comprehensive estimate of damages related to impact cascades remains difficult to obtain.

How then should tracking, monitoring, forecasting, and understanding the spatial extent and temporal relationships of cascading impacts be achieved? A global monitoring system that tracks these cascades would help develop a database resource for critically examining and better understanding such events. This information would inform risk management and draw attention to otherwise overlooked feedback processes. The data could also facilitate the development of more robust information that provides insights on natural disasters and consequences at global scales. Two-way coupling of models in which information is exchanged through many iterations could promote better understanding of interactions among hazards, human activities, and their multisectoral effects. Ultimately, considering impact cascades, feedback loops, and their complex interactions should help direct preparations for future severe events and rapid transitions or whiplash between climatic extremes. □

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IMMUNOLOGY

Designed to remember

Specific DNA sequence features encode the persistence of epigenetic memory of inflammation

Guillaume Blot[1] and **Przemyslaw Sapieha**[1,2]

Inflammation is transient, but its consequences can be lifelong. Even brief episodes of inflammation can lead to long-lasting changes in chromatin that control gene expression. These changes can poise cells to respond differently to subsequent inflammation episodes. For example, an initial encounter with environmental stressors, such as bacteria, viruses, or Western diets, durably tunes innate immune responses to later (related or unrelated) challenges (*1*, *2*). These mechanisms also exist in nonimmune cells, including the epidermal cells of human skin (*3*, *4*). Why some inflammation-induced chromatin alterations are transient whereas others persist is unclear. On page 1347 of this issue, Cowley *et al.* (*5*) report that specific regions dense in cytosine and guanosine dinucleotides (CpG-rich) are epigenetically modified during inflammation to enable gene expression and that these changes persist during the animal's lifetime. The finding has implications for understanding how the genome determines the longevity of memory, which affects tissue fitness.

Epigenetic alterations to chromatin (the architecture of DNA wrapped around histone proteins) affect gene expression without modifying the DNA sequence. These modifications can change the conformation of chromatin between open (accessible to transcription factors) and closed states. Cowley *et al.* applied imiquimod to the skin of mice to induce psoriasis-like inflammation, which self-resolves within 30 days. This transient perturbation allowed the authors to distinguish between chromatin changes that fade during the resolution of inflammation and those that persist long after inflammation has ended. Cowley *et al.* defined "long-term memory domains" as genomic regions that were more open in inflamed skin compared with age-matched naïve skin at day 30 and whose genomic accessibility persisted 1 year after inflammation. Notably, most of these regions remained open for up to 2 years (a common lifetime for a mouse) after inflammation. Conversely, "short-term memory domains" closed up between day 30 and year 1. Single-cell profiling revealed that long-lasting genomic memory is broadly distributed across the epidermal stem cell population rather than being confined to a niche-restricted subset.

The persistence of epigenetic alterations has been primarily explained through mechanisms that regulate gene expression. Chromatin states induced by a specific stimulus are sustained after its withdrawal through transcription factor binding, cofactor recruitment, and chromatin remodeling, which are reinforced by feedback loops within regulatory circuits (*6*, *7*). In this framework, the genome specifies where regulators can act, whereas the longevity of epigenetic alterations results chiefly from how long these regulatory processes can maintain a chromatin configuration that permits transcription. Cowley *et al.* refined the hierarchy of mechanisms underlying inflammatory memory by parsing out establishment from persistence. During the initial inflammatory response in the mouse model of psoriasis, the protein signal transducer and activator of transcription 3 (STAT3) and the activator protein–1 (AP-1) transcription factor helped to open memory domains. This was accompanied by methylation and acetylation of histone H3 (H3K4me1 and H3K27ac epigenetic alterations, respectively) in those memory domains. However, these mechanisms do not explain the longevity of inflammatory memory because some of these open domains subsequently closed, whereas others remained accessible for years.

Instead, Cowley *et al.* noted that longevity was associated with features of the DNA sequence itself (see the figure). Long-term memory

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